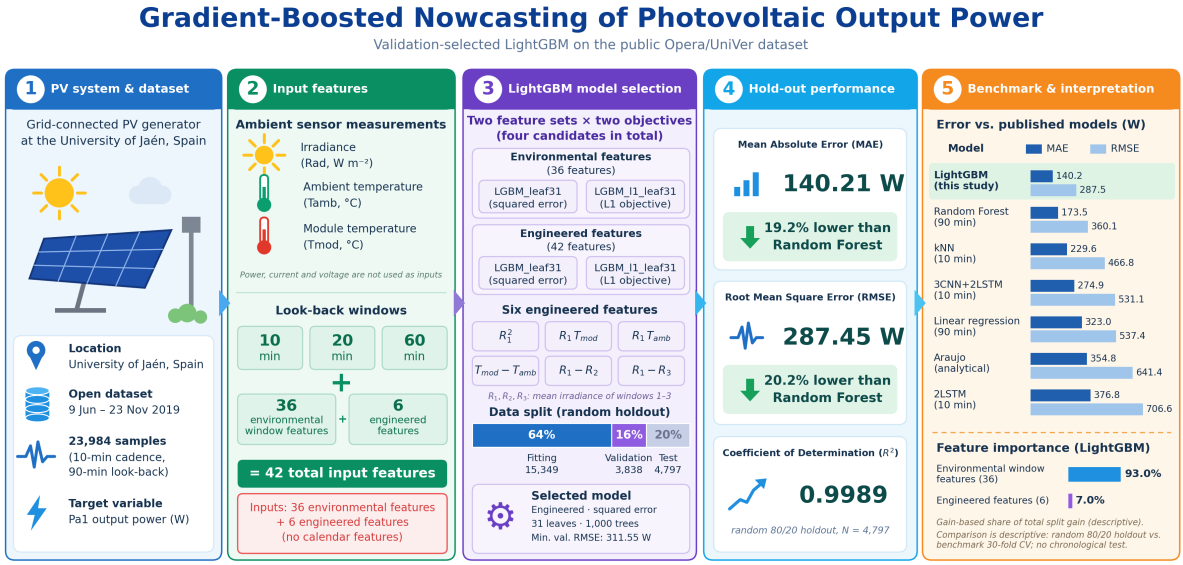
Supplementary Material

Photovoltaic Power Nowcasting Using Ambient Sensor Data:
An Evaluation of LightGBM on the OPERA/UniVer Benchmark

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Companion to the main manuscript. Supplementary section, table, figure and equation numbers carry the prefix S.



Dataset, LightGBM model, evaluation, and comparison with the published benchmark.

S1 Inputs, Alignment, and Sample Counts

Input files. The analysis uses the benchmark files `opera_dataset_1w_10min.csv` (24,031 rows) and `opera_dataset_3w_90min.csv` (24,024 rows). Each sample has environmental statistics for irradiance (`Rad`), ambient temperature (`Tamb`) and module temperature (`Tmod`) over three incremental windows (10, 20 and 60 minutes). Each signal contributes its mean, maximum, minimum and standard deviation in each window, giving

$$d_{\text{env}} = 3 \text{ signals} \times 4 \text{ statistics} \times 3 \text{ windows} = 36. \tag{S1}$$

The prediction target is `Pa1`. Predictors whose names indicate output power, current, voltage, generation, production or energy are excluded.

Alignment. Rows in the 90-minute feature file are retained only when the nine trailing 10-minute timestamps, $t, t-10, \ldots, t-80$ minutes, are present and consecutive in the 10-minute file. The aligned dataset contains $n = 23{,}984$ observations spanning 9 June to 23 November 2019. At the 24,024 timestamps shared by the source files, the `Pa1` values agree (maximum absolute difference 0 W). The retained predictors contain no missing values. The target ranges from 0 to 28,472.31 W; its mean is 6,700.64 W and its median is 552.24 W. About 26.4% of aligned observations are below 1 W, corresponding mainly to night-time operation; these observations are retained.

Data splits. With scikit-learn's shuffled split convention and fixed seeds, the outer split uses seed 42 and holds out 20% of the aligned observations for testing. The remaining training set is then split with seed 43

to form the fitting and validation subsets:

$$n_{\text{test}} = \lceil 0.20 \times 23{,}984 \rceil = 4{,}797, \qquad n_{\text{train}} = 23{,}984 - 4{,}797 = 19{,}187, \tag{S2}$$

$$n_{\text{val}} = \lceil 0.20 \times 19{,}187 \rceil = 3{,}838, \qquad n_{\text{fit}} = 19{,}187 - 3{,}838 = 15{,}349. \tag{S3}$$

Figure S1 summarises the implemented pipeline.

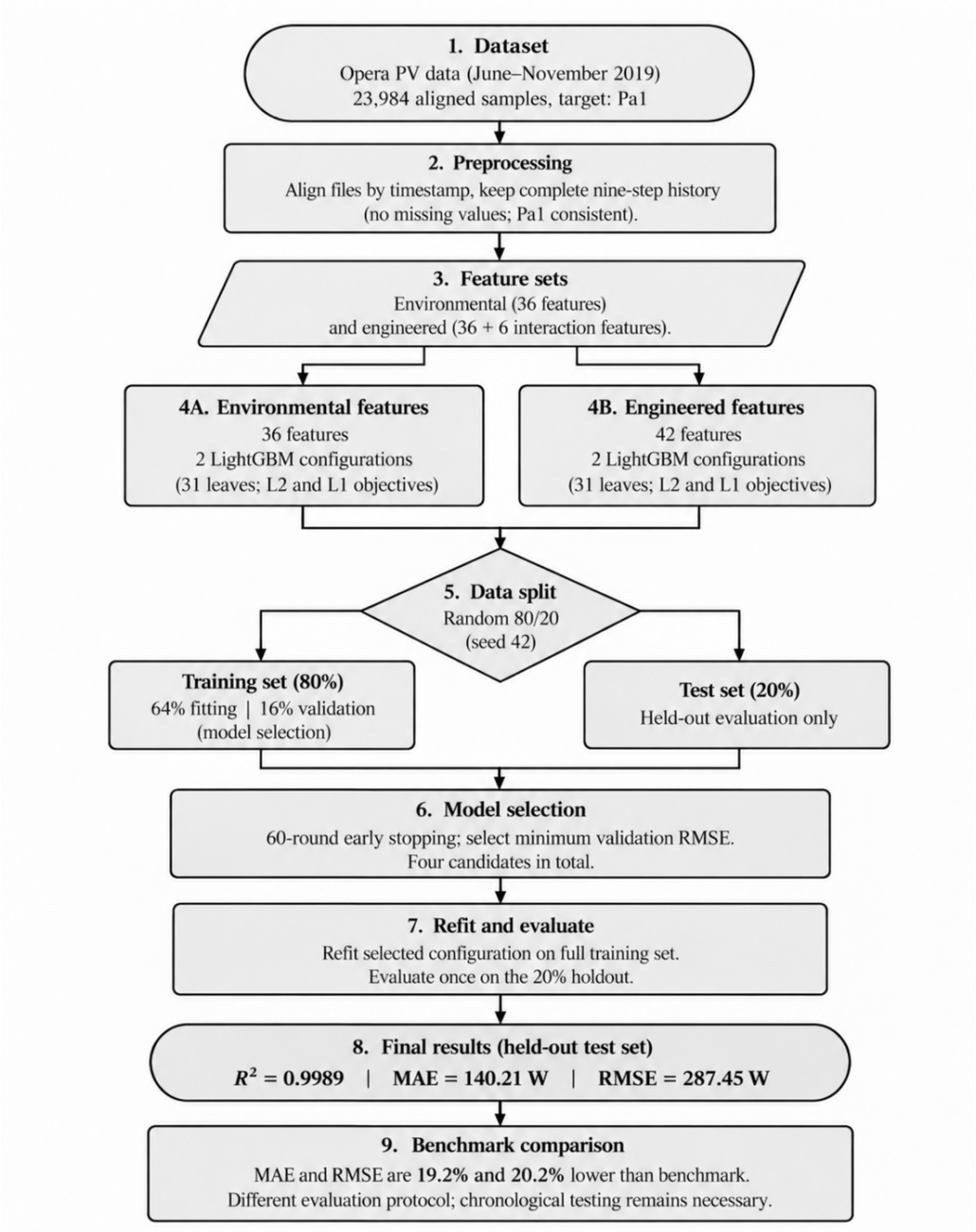


Figure S1: Implemented pipeline for photovoltaic power nowcasting with LightGBM. The feature sets contain 36 environmental variables, or those 36 variables plus six engineered features. Four configurations are compared by validation RMSE, and the selected model is evaluated once on the random holdout.

S2 Feature Construction

The environmental-only feature set contains 36 predictors. The engineered set adds six deterministic transformations of measured environmental quantities, giving 42 predictors in total. No timestamp-derived, calendar or cyclic time features are used.

Let R_1, $T_{m,1}$ and $T_{a,1}$ denote the first-window mean irradiance, module temperature and ambient temperature, respectively, and let R_2 and R_3 be the mean irradiance in the second and third windows. The additional features are

$$x_{37} = R_1^2, \qquad x_{38} = R_1 T_{m,1}, \qquad x_{39} = R_1 T_{a,1}, \tag{S4}$$

$$x_{40} = T_{m,1} - T_{a,1}, \qquad x_{41} = R_1 - R_2, \qquad x_{42} = R_1 - R_3. \tag{S5}$$

The first three terms allow nonlinear irradiance and irradiance–temperature interactions; the temperature difference represents the module-to-ambient temperature gap, and the two irradiance differences describe changes across the nested windows. These features reuse existing measurements and do not add observations. In the implementation the engineered matrix is stored as 32-bit floats, whereas the environmental-only matrix keeps 64-bit floats.

S3 LightGBM Model and Training Configuration

S3.1 Gradient-boosting formulation

Gradient boosting builds an additive predictor of the form

$$F_m(\mathbf{x}) = F_{m-1}(\mathbf{x}) + \nu h_m(\mathbf{x}), \qquad r_{im} = -\left.\frac{\partial L(y_i, F(\mathbf{x}_i))}{\partial F(\mathbf{x}_i)}\right|_{F=F_{m-1}}, \tag{S6}$$

where h_m is the regression tree added at iteration m, ν is the learning rate, and r_{im} is the negative loss gradient at the current predictor. For a twice-differentiable loss, let g_i and q_i denote the first and second derivatives of the loss with respect to the prediction, evaluated at the current predictor. Under the usual second-order tree approximation, a tree with J leaves, leaf weights w_j and regularisation strengths λ_1 and λ_2 has the objective

$$\mathcal{L}_m \simeq \sum_{j=1}^{J} \left[G_j w_j + \frac{1}{2}(H_j + \lambda_2) w_j^2 + \lambda_1 |w_j| \right], \quad G_j = \sum_{i \in I_j} g_i, \quad H_j = \sum_{i \in I_j} q_i, \tag{S7}$$

where I_j contains the observations assigned to leaf j. Minimising the objective gives

$$w_j^* = -\frac{\mathcal{T}_{\lambda_1}(G_j)}{H_j + \lambda_2}, \quad \mathcal{T}_{\lambda_1}(G) = \operatorname{sign}(G) \max(|G| - \lambda_1, 0), \tag{S8}$$

and the corresponding split gain, for a node with left and right children L and R and totals $G = G_L + G_R$ and $H = H_L + H_R$, is

$$\text{Gain} = \frac{1}{2}\left[\frac{\mathcal{T}_{\lambda_1}(G_L)^2}{H_L + \lambda_2} + \frac{\mathcal{T}_{\lambda_1}(G_R)^2}{H_R + \lambda_2} - \frac{\mathcal{T}_{\lambda_1}(G)^2}{H + \lambda_2} \right]. \tag{S9}$$

LightGBM uses histogram-based split finding and leaf-wise growth. For squared-error loss the gradient is $F_{m-1}(\mathbf{x}_i) - y_i$ and the second derivative is 1, so the expressions above apply directly. The absolute-error loss is not twice differentiable; for the `regression_l1` objective LightGBM uses the sign of the residual as the gradient and sets leaf values from residual quantiles, so the second-order expressions above describe only the squared-error candidates.

S3.2 Candidate configurations

The code compares two feature sets (environmental-only and engineered) with two objectives (squared error and absolute error), giving four candidates. The shared model settings are listed in Table S1.

Table S1: LightGBM settings used for all four candidates, apart from the feature set and objective.

Parameter	**Value**
Objective	`regression` (squared error) or `regression_l1` (absolute error)
Maximum number of trees	1,000
Learning rate ν	0.025
Number of leaves	31
Maximum depth	-1 (no explicit depth limit)
Minimum samples per leaf	25
Row subsampling	0.90, with `subsample_freq`=1
Column subsampling	0.90
L_2 regularisation (λ_2)	3.0
L_1 regularisation (λ_1)	0.05
Random seed	42
Early-stopping patience	60 rounds on the validation subset
Early-stopping metric	LightGBM default for the objective (squared error or absolute error)

Each candidate is fitted on the 15,349-row fitting subset and evaluated on the 3,838-row validation subset for early stopping. The candidate results are computed at each model's reported best iteration. The selected configuration is then refitted on the full 19,187-row outer training set using that iteration count and evaluated once on the 4,797-row test set. The selected candidate reached the configured maximum of 1,000 trees; no test labels were used for model selection.

S4 Evaluation Metrics and Benchmark Calculations

For test observations y_i, predictions $\hat{y}_i$, residuals $e_i = y_i - \hat{y}_i$ and $N = 4{,}797$, the reported metrics are

$$\mathrm{MAE} = \frac{1}{N}\sum_{i=1}^{N}|e_i|, \qquad \mathrm{MSE} = \frac{1}{N}\sum_{i=1}^{N}e_i^2, \qquad \mathrm{RMSE} = \sqrt{\mathrm{MSE}}, \tag{S10}$$

$$\mathrm{SSE} = \sum_{i=1}^{N}e_i^2, \qquad \mathrm{SST} = \sum_{i=1}^{N}(y_i - \bar{y})^2, \qquad R^2 = 1 - \frac{\mathrm{SSE}}{\mathrm{SST}}, \tag{S11}$$

where $\bar{y}$ is the mean measured test power. With $s_y^2 = \mathrm{SST}/N$, this can also be written as $R^2 = 1 - \mathrm{MSE}/s_y^2$. Jensen's inequality implies $\mathrm{MAE} \le \mathrm{RMSE}$. For zero-mean Gaussian errors, $\mathrm{MAE}/\mathrm{RMSE} \approx \sqrt{2/\pi} = 0.798$; the observed ratio is $140.212/287.451 \approx 0.488$, and the largest absolute error is 4,674.31 W. This ratio alone is not a formal test of the residual distribution.

Worked test-set values. The values below are recomputed from `random_test_predictions.csv`, using its `actual_Pa1_W` and `selected_LightGBM_prediction_W` columns:

$$\sum_i |e_i| = 672{,}597.1366\ \mathrm{W}, \qquad \mathrm{MAE} = \frac{672{,}597.1366}{4{,}797} = 140.2120\ \mathrm{W}, \tag{S12}$$

$$\mathrm{SSE} = 396{,}367{,}799.4586\ \mathrm{W}^2, \qquad \mathrm{RMSE} = \sqrt{\frac{396{,}367{,}799.4586}{4{,}797}} = 287.4513\ \mathrm{W}, \tag{S13}$$

$$\mathrm{SST} = 369{,}969{,}776{,}817.0953\ \mathrm{W}^2, \qquad R^2 = 1 - \frac{396{,}367{,}799.4586}{369{,}969{,}776{,}817.0953} = 0.99892865. \tag{S14}$$

The test-target variance with denominator N is $s_y^2 = 77{,}125{,}240.1120\ \mathrm{W}^2$, corresponding to $s_y = 8{,}782.10$ W. The absolute error is below 100 W for 63.52% of test observations and below 500 W for 93.87%.

Percentage reduction relative to a published error. For a reference error m_{ref} and the present test error m, define

$$\Delta(m) = 100\,\frac{m_{\text{ref}} - m}{m_{\text{ref}}}; \qquad \Delta(m) > 0 \text{ indicates a lower reported error.} \tag{S15}$$

Relative to the benchmark Random Forest (90-minute window; MAE 173.47 W, RMSE 360.13 W), LightGBM reports reductions of 19.17% and 20.18%. Relative to the Araujo analytical model (MAE 354.81 W, RMSE 641.36 W), the reductions are 60.48% and 55.18%. Relative to the 3CNN+2LSTM model using 10-minute segmentation (MAE 274.87 W, RMSE 531.08 W), the reductions are 48.99% and 45.87%. These calculations are descriptive: the published results were obtained under a 30-fold cross-validation protocol, whereas the present result is from one random holdout, so the comparison is not protocol-matched.

Table S2: Consistency checks between the main manuscript and the recomputed outputs.

Quantity	**Main manuscript**	**Recomputed value**	**Source**
Outer training / test rows	19,187 / 4,797	19,187 / 4,797	Split
Feature counts (environmental / engineered total)	36 / 42	36 / 42	Features
Holdout MAE / RMSE	140.21 / 287.45 W	140.2120 / 287.4513 W	Predictions
Holdout R^2	0.998929	0.99892865	Predictions
Reduction vs. Random Forest (MAE / RMSE)	19.2 / 20.2%	19.17 / 20.18%	Benchmark
Reduction vs. Araujo (MAE / RMSE)	60.5 / 55.2%	60.48 / 55.18%	Benchmark
Reduction vs. 3CNN+2LSTM (MAE / RMSE)	49.0 / 45.9%	48.99 / 45.87%	Benchmark
Absolute error below 100 / 500 W	63.5 / 93.9%	63.52 / 93.87%	Predictions
Largest absolute error	4,674.31 W	4,674.3135 W	Predictions

S5 Validation Results and Model Selection

The four candidate configurations are ranked by validation RMSE in watts. The selection rule is

$$c^* = \arg\min_{c \in \mathcal{C}} \text{RMSE}_{\text{val}}(c), \tag{S16}$$

where c is a candidate configuration and $\mathcal{C}$ contains the two feature sets crossed with the two objectives. Table S3 is sorted by validation RMSE; the selected candidate is `env_engineered__LGBM_leaf31`.

Table S3: Validation metrics for the four LightGBM candidates. Early stopping uses the validation subset; the selected candidate minimises validation RMSE.

Feature set	**Objective**	R^2_{val}	**MAE (W)**	**RMSE (W)**	**Best iteration**	**Selected**
Engineered (42)	**Squared error**	**0.998725**	**147.31**	**311.55**	**1000**	**Yes**
Environmental-only (36)	Squared error	0.998684	151.61	316.48	996	No
Engineered (42)	Absolute error	0.998408	151.57	348.03	1000	No
Environmental-only (36)	Absolute error	0.998275	155.57	362.28	1000	No

Using squared-error loss, the six engineered features reduce validation MAE by 2.84% and RMSE by 1.56% relative to the environmental-only set. Using absolute-error loss, the corresponding reductions are 2.57% and 3.93%. These values are validation comparisons, not independent test estimates; the test performance of the environmental-only candidates was not computed separately.

For the selected model, gain-based feature importance assigns 92.98% of total split gain to the 36 environmental window features and 7.02% to the six engineered features (rounded to 93.0% and 7.0%). The largest individual contribution is from `Rad_avg`. Gain importance describes how the fitted trees used the features; it does not establish causal effects and does not replace a feature-ablation experiment.

S6 Reproduction Procedure

1. Load both Opera CSV files; validate timestamps and target values, check target consistency on overlapping timestamps, and retain rows with complete consecutive 10-minute history ($n = 23{,}984$).
2. Build the 36 environmental features and the six engineered features described in Section S2.
3. Create the outer training/test split with seed 42 (19,187 / 4,797 rows), then the fitting/validation split with seed 43 (15,349 / 3,838 rows).
4. Fit each of the four candidate models using the shared settings in Table S1, with early stopping (patience 60 rounds) on the validation subset. Record validation R^2, MAE, RMSE and best iteration.
5. Select the candidate with the lowest validation RMSE, refit it on all 19,187 outer-training rows using its selected iteration count, and evaluate once on the untouched test set.
6. Save the validation table, test predictions, comparison metrics, selected-model feature importance and run recipe. In the reported run these files are `random_validation_search.csv`, `random_test_predictions.csv`, `random_comparison.csv`, `random_selected_feature_importance.csv` and `random_recipe.json`.

S7 Methodological Checklist

- *Applied:* complete-window alignment; predictors exclude output power, current, voltage and target-derived variables.
- *Applied:* 36 environmental predictors or 42 engineered predictors (no calendar features); four candidates compared and selected by validation RMSE.
- *Applied:* refit of the selected candidate on all outer-training rows and a single evaluation on the untouched holdout.
- *Not applied:* chronological or blocked evaluation, nested selection, repeated seeds, confidence intervals, test-set ablation, or independent-site or season validation.

S8 Notation

Table S4: Notation used in the supplementary derivations.

Symbol	Meaning	Symbol	Meaning
`Pa1`	Measured output power and prediction target, W	h_m	Regression tree added at iteration m
R_k	Mean irradiance in window k	$T_{a,1}$, $T_{m,1}$	First-window ambient and module temperature
y_i, $\hat{y}_i$, e_i	Measurement, prediction and residual	g_i, q_i	First and second loss derivatives where defined
G_j, H_j	Sums of g_i and q_i over the observations in leaf j	λ_1, λ_2	L_1 and L_2 regularisation strengths
N, n	Test rows (4,797); aligned rows (23,984)	MAE, RMSE	Mean absolute error; root mean squared error
SSE, SST	Residual and total sums of squares	s_y^2	Test-target variance with denominator N
$\mathcal{C}$, c, c^*	Candidate set, a candidate, the selected candidate	$\Delta(m)$	Percentage reduction relative to a reference error
ν	Learning rate	R^2	Coefficient of determination